Lab Exercise 17- Creating Secrets in Kubernetes

Creating secrets in Kubernetes is an essential practice for managing sensitive information such as passwords, OAuth tokens, and SSH keys. Let's go through a lab exercise that demonstrates how to create different types of secrets in Kubernetes.

Step 1: Create a Generic Secret

Create a secret from literal values:

```
kubectl create secret generic my-secret --from-literal=username=admin --from-literal=password='P@ssword'
```

```
devanksilswal@devanks-MacBook-Air ex_s % kubectl create secret generic my-secret --from-literal=username=admin --from-literal=password='P@ssw0rd'
secret/my-secret created
```

Verify the secret:

```
kubectl get secrets my-secret -o yaml
```

```
devanksilswal@devanks-MacBook-Air ex_s % kubectl get secrets my-secret -o yaml
apiVersion: v1
data:
  password: UEBzc3cwcmQ=
  username: YWRtaW4=
kind: Secret
metadata:
  creationTimestamp: "2026-04-17T12:03:56Z"
  name: my-secret
  namespace: my-namespace
  resourceVersion: "185243"
  uid: 83cceccd-c06f-4716-8b10-570f363feb94
type: Opaque
```

The output will be base64 encoded. To decode, you can use:

```
echo -n “P@ssword” | base64
```

```
devanksilswal@devanks-MacBook-Air ex_s % echo -n "P@ssw0rd" | base64
UEBzc3cwcmQ=
```

Step 2: Create a Secret from a File

Create a file with your sensitive data:

```
echo -n 'my-username' > ./username.txt



echo -n 'my-password' > ./password.txt
```

```
devanksilswal@devanks-MacBook-Air ex_s % echo -n 'my-username' > username.txt
echo -n 'my-password' > password.txt
```

Create a secret from the file:

```
kubectl create secret generic my-file-secret --from-file=username=./username.txt --from-file=password=./password.txt
```

```
devanksilswal@devanks-MacBook-Air ex_s % kubectl create secret generic my-file-secret --from-file=username=./username.tx
t --from-file=password=./password.txt
secret/my-file-secret created
```

Verify the secret:

```
kubectl get secrets my-file-secret -o yaml
```

```
devanksilswal@devanks-MacBook-Air ex_s % kubectl get secrets my-file-secret -o yaml
apiVersion: v1
data:
  password: bXktcGFzc3dvcmQ=
  username: bXktdXNlcm5hbWU=
kind: Secret
metadata:
  creationTimestamp: "2026-04-17T12:07:38Z"
  name: my-file-secret
  namespace: my-namespace
  resourceVersion: "185647"
  uid: 1c211622-542b-45c4-b979-e717ecf8e016
type: Opaque
```

Step 3: Create a Docker Registry Secret

Create a Docker registry secret:

```
kubectl create secret docker-registry my-registry-secret --docker-username=<your-username> --docker-password=<your-password> --docker-email=<your-email> --docker-server=<your-docker-server>
```

```
devanksilswal@devanks-MacBook-Air ex_s % kubectl create secret docker-registry my-registry-secret \
--docker-username=devank \
--docker-password=123456 \
--docker-email=devanksilswal22@gmail.com \
--docker-server=https://index.docker.io/v1/
secret/my-registry-secret created
```

Verify the secret:

```
kubectl get secrets my-registry-secret -o yaml
```

```
devanksilswal@devanks-MacBook-Air ex_s % kubectl get secrets my-registry-secret -o yaml
apiVersion: v1
data:
  .dockerconfigjson: eyJhdXRocyI6eyJodHRwczovL2luZGV4LmRvY2tlci5pby92MS8iOnsidXNlcm5hbWUiOiJkZXZhbmsiLCJwYXNzd29yZCI6IjE
yMzQ1NiIsImVtYWlsIjoiZGV2YW5rc2lsc3dhbDIyQGdtYWlsLmNvbSIsImF1dGgiOiJaR1YyWVc1ck9qRXlNelExTmc9PSJ9fX0=
kind: Secret
metadata:
  creationTimestamp: "2026-04-17T12:19:30Z"
  name: my-registry-secret
  namespace: my-namespace
  resourceVersion: "186414"
  uid: 193cae88-ec82-4d3c-80e1-566a6502415a
type: kubernetes.io/dockerconfigjson
```

Step 4: Create a Secret Using a YAML Manifest

Create a YAML file for the secret:

Create a file named **my-secret.yaml** with the following content:

```
apiVersion: v1
kind: Secret
metadata:
  name: my-manifest-secret
type: Opaque
data:
  username: YWRtaW4=    # base64 encoded value of 'admin'
  password: UEBzc3cwcmQ=  # base64 encoded value of 'P@ssw0rd'
```

Apply the YAML file:

```
kubectl apply -f my-manifest-secret.yaml
```

```
devanksilswal@devanks-MacBook-Air ex_s % touch my-manifest-secret.yaml
devanksilswal@devanks-MacBook-Air ex_s % nano my-manifest-secret.yaml
devanksilswal@devanks-MacBook-Air ex_s % kubectl apply -f my-manifest-secret.yaml
secret/my-manifest-secret created
```

Verify the secret:

```
kubectl get secrets my-manifest-secret -o yaml
```

```
devanksilswal@devanks-MacBook-Air ex_s % kubectl get secrets my-manifest-secret -o yaml
apiVersion: v1
data:
  password: UEBzc3cwcmQ=
  username: YWRtaW4=
kind: Secret
metadata:
  annotations:
    kubectl.kubernetes.io/last-applied-configuration: |
      {"apiVersion":"v1","data":{"password":"UEBzc3cwcmQ=","username":"YWRtaW4="},"kind":"Secret","metadata":{"annotatio
ns":{},"name":"my-manifest-secret","namespace":"my-namespace"},"type":"Opaque"}
  creationTimestamp: "2026-04-17T12:24:03Z"
  name: my-manifest-secret
  namespace: my-namespace
  resourceVersion: "186914"
  uid: 35ec2a58-9f9e-4838-a843-44f001202473
type: Opaque
```

Step 5: Use Secrets in Pods

Create a pod that uses the secret:

Create a file named **pod-with-secret.yaml** with the following content:

```
apiVersion: v1
kind: Pod
metadata:
  name: pod-with-secret
spec:
  containers:
  - name: mycontainer
    image: nginx
    env:
    - name: USERNAME
      valueFrom:
        secretKeyRef:
          name: my-secret
          key: username
    - name: PASSWORD
      valueFrom:
        secretKeyRef:
          name: my-secret
          key: password
```

Apply the YAML file:

```
kubectl apply -f pod-with-secret.yaml
```

```
devanksilswal@devanks-MacBook-Air ex_s % touch pod-with-secret.yaml
devanksilswal@devanks-MacBook-Air ex_s % nano pod-with-secret.yaml
devanksilswal@devanks-MacBook-Air ex_s % kubectl apply -f pod-with-secret.yaml
pod/pod-with-secret created
```

Verify the pod is running and the secret is being used:

```
kubectl exec pod-with-secret -- printenv | grep -E 'USERNAME|PASSWORD'
```

```
devanksilswal@devanks-MacBook-Air ex_s % kubectl exec pod-with-secret -- printenv | grep -E 'USERNAME|PASSWORD'
USERNAME=admin
PASSWORD=P@ssw0rd
```

Step 7: Clean Up

Delete the secrets:

```
kubectl delete secret my-secret
kubectl delete secret my-file-secret
kubectl delete secret my-registry-secret
kubectl delete secret my-manifest-secret
```

Delete the pod:

```
kubectl delete pod pod-with-secret
```

```
devanksilswal@devanks-MacBook-Air ex_s % kubectl delete secret my-secret
kubectl delete secret my-file-secret
kubectl delete secret my-registry-secret
kubectl delete secret my-manifest-secret
secret "my-secret" deleted from my-namespace namespace
secret "my-file-secret" deleted from my-namespace namespace
secret "my-registry-secret" deleted from my-namespace namespace
secret "my-manifest-secret" deleted from my-namespace namespace
devanksilswal@devanks-MacBook-Air ex_s % kubectl delete pod pod-with-secret
pod "pod-with-secret" deleted from my-namespace namespace
```